GREEN TECH HACKATHON 2026 · ORACLE ACADEMY × STRATHMORE iLABAFRICA

HarvestLink

Hackathon Readiness & Execution Plan

Team	Bora Chakula
Institution	Murang'a University of Technology
Product	HarvestLink: demand-first agricultural marketplace
Primary track	5. Water Stewardship & Climate-Smart Agriculture (secondary: 4. Sustainable Transport & Logistics)
Prepared	Wednesday, 14 October 2026
Status	Draft for team review

1. Executive Summary

HarvestLink connects farmers with verified buyers before harvest. Buyers post upcoming produce needs. The platform then matches and aggregates smallholder farmers by quantity, quality grade, timing and location, coordinates shared logistics, and routes lower grades to suitable markets. The result is better market access for farmers, less post-harvest loss, fewer empty or duplicate transport trips, and less food waste.

Bora Chakula is one of 12 finalists. This plan covers the next 48 hours: building a working prototype on Oracle APEX and OCI, gathering first-hand evidence from farmers in local markets, submitting by Thursday 15 October at 4:00 PM, and delivering a winning pitch on Friday 16 October.

2. Key Dates & Deadlines

Date / Time	Milestone
Wed 14 Oct	Bootcamp Day 3: Oracle Cloud training; mentor sessions. Build starts in earnest.
Thu 15 Oct, 6:00–10:00 AM	Market field interviews (team field unit).
Thu 15 Oct, 9:00 AM	Daily check-in (Zoom).
Thu 15 Oct, 4:00 PM	**SUBMISSION DEADLINE: email ilabee@strathmore.edu**
Fri 16 Oct, 8:00 AM	Arrival at Strathmore University (Gate B, Strathmore Business School, Keri Campus).
Fri 16 Oct, 9:30 AM	Pitches begin: 7 min pitch + 3 min Q&A per team.
Fri 16 Oct, 12:00 PM	Awards ceremony (top 3 shortlisted at 11:30 AM).

Friday reminders: carry your National ID; smart casual dress; parking is limited, so carpool or use a taxi.

3. What the Judges Expect

The Grand Jury: Billy Otieno (Enterprise AI & Cloud Solutions Architect), Lavynne Ayisi (Legal Counsel) and Albert Bwire (Principal Support Account Manager), all from Oracle. Their checklist rewards Oracle technology adoption and measurable sustainability impact.

3.1 Submission Checklist

Area	Requirement	Done
Project overview	Team name, institution, team members with roles Hackathon track and project title Short description (max 200 words)	☐
Technical	GitHub repository link App deployed on OCI (APEX workspace on OCI or APEX on-premise) Uses Oracle technology; judges can view the app	☐
Impact & innovation	Problem statement Solution & approach Innovation: what makes it unique Sustainability impact on the green economy	☐
Pitch	Max 5 slides (problem, solution, tech stack, impact) Prototype link	☐
Final	Documentation clear and concise Sent by Thu 15 Oct, 4:00 PM to ilabee@strathmore.edu	☐

4. Positioning & Impact Story

4.1 Problem

- Farmers harvest first and look for buyers later, so they sell under pressure to brokers at low prices or watch produce spoil.
- Smallholders each transport small volumes separately, wasting fuel and adding emissions.
- Lower-grade produce has no planned outlet and is often discarded.

4.2 What Makes HarvestLink Different

- **Demand-first:** buyer commitments exist before harvest, so farmers plan to meet real demand.
- **Aggregation:** many small farmers combine to fill one large order that none could fill alone.
- **Grade routing:** every grade has a destination: premium buyers, local markets, processors or schools.
- **Logistics clustering:** one planned pickup route replaces many individual trips.

4.3 Impact Metrics We Will Show

Metric	How it is calculated
Post-harvest loss avoided (kg)	Matched volume × baseline loss rate (from farmer interviews)

Metric	How it is calculated
Transport distance saved (km)	Sum of individual farmer trips − aggregated route distance
CO_2 avoided (kg)	Distance saved × truck emission factor (kg CO_2 per km)
Farmer income uplift (KES)	(Direct buyer price − broker price) × volume sold
Farmers reached	Count of farmers with confirmed matches

5. Technical Approach

We will build on Oracle's stack. This meets the "Adoption of Oracle Technology" criterion and is the fastest route to a reliable, hosted demo.

Component	Choice
Hosting	Oracle Cloud Infrastructure (Always Free tier via Oracle Academy Cloud Program)
Database	Oracle Autonomous Database (farmers, harvests, buyers, demands, matches, routes)
Application	Oracle APEX: buyer and farmer forms, matching results, map, impact dashboard
Matching engine	PL/SQL scoring function: quantity fit, grade match, harvest-window overlap and distance, with a "why matched" explanation
AI (stretch)	APEX AI / Select AI: buyers describe a demand in plain language and it is turned into a structured request
Maps	APEX Map region: farmer pins, buyer location, aggregated pickup cluster
Source control	GitHub repo BoraChakula: APEX export, SQL scripts, seed data, README

6. MVP Scope (Demo Flow)

Build only what will be demonstrated. Everything else goes on the roadmap slide.

1. **Buyer posts demand:** e.g. 2 tonnes Grade A tomatoes, Nairobi, deliver 28 October.
2. **Farmer registers expected harvest:** crop, estimated quantity, grade, harvest date, location. Mobile-friendly.
3. **Match & aggregate:** ranked farmers combined to fill the order, with reasons shown.
4. **Grade routing:** Grade B/C volumes routed to processors, schools or local markets.
5. **Logistics view:** map of pickup cluster with km and CO_2 saved versus individual trips.
6. **Impact dashboard:** kg matched, waste avoided, CO_2 saved, farmers served.

Seed data: realistic Murang'a and Central Kenya crops (tomatoes, avocados, bananas, cabbages, maize), using figures gathered in the market interviews.

7. Field Research: Market Interviews

7.1 Logistics

When	Thursday 15 October, 6:00–10:00 AM (markets are busiest early)
Where	Murang'a town market (alternative: Wakulima/Marikiti market, Nairobi)
Who	1–2 team members; the rest of the team keeps building
Target	5–10 farmers and 2–3 traders or brokers
Bring	Printed question sheet (Appendix A), charged phone for photos and video, notebook, student ID

7.2 Farmer Questions

#	English	Kiswahili
1	What do you grow, and how much per season?	Unalima nini, na unavuna kiasi gani kwa msimu?
2	Do you know who will buy your harvest before you harvest? How do you find buyers?	Je, unajua mnunuzi wa mavuno yako kabla ya kuvuna? Unapataje wanunuzi?
3	How much of your harvest is lost or spoiled? Why?	Ni kiasi gani cha mavuno yako kinapotea au kuharibika? Kwa nini?
4	How do you get produce to market? Cost per trip and distance?	Unafikishaje mazao sokoni? Gharama ya safari ni kiasi gani, na ni umbali gani?
5	What do brokers pay you compared with the market price?	Madalali wanakulipa kiasi gani ukilinganisha na bei ya soko?
6	What happens to lower-grade or damaged produce?	Mazao ya daraja la chini au yaliyoharibika huishia wapi?
7	Would you join other farmers to fill a big order and share transport?	Ungekubali kujiunga na wakulima wengine kujaza oda kubwa na kugawana usafiri?
8	What phone do you use? Do you use M-Pesa or WhatsApp?	Unatumia simu ya aina gani? Unatumia M-Pesa au WhatsApp?

7.3 Trader / Broker Questions

- Where do you source produce, and how far in advance do you know your volumes?
- How often can you not find enough supply, or end up with too much?
- Would you commit to an order before harvest if quality and delivery were guaranteed?

7.4 Ethics & Data Capture

- Introduce yourselves as students; explain the purpose; ask permission before recording or photographing.
- Do not record personal details beyond first name and location unless the person agrees.
- Capture 1–2 strong quotes, key numbers (loss %, broker margin, transport cost) and short consented testimonials.

- Share findings with the team by 11:00 AM Thursday for the deck and seed data.

8. Pitch Deck (5 Slides, 7 Minutes)

Slide	Title	Content	Time
1	The Problem	Farmer quote and photo; % harvest lost; broker squeeze	1:00
2	Our Solution	Demand-first matching, aggregation, grade routing	1:00
3	Live Demo	Buyer posts demand → match → route → impact	3:00
4	Built on Oracle	APEX, Autonomous DB, OCI; matching logic; architecture	1:00
5	Impact & Next Steps	Waste and CO_2 avoided; Murang'a pilot; business model	1:00

8.1 Anticipated Judge Questions

Question	Prepared answer (outline)
How do you protect farmer and buyer data?	Minimal data collected, consent at sign-up, role-based access in APEX, Kenya Data Protection Act 2019 compliance
How are buyers verified?	KRA PIN / business registration check, references, order history and ratings
What about farmers without smartphones?	Roadmap: USSD/SMS registration; cooperative and extension-officer agents
How do you make money?	Small transaction fee on matched orders, charged to buyers; premium logistics services
How does the AI matching work?	Weighted scoring on quantity, grade, timing and distance; explainable results; ML added as data grows
Why would brokers or farmers trust it?	Pre-agreed prices, M-Pesa payment on delivery, cooperative partnerships

9. Timeline & Owners

When	Task	Owner
Wed afternoon	Set up OCI tenancy and APEX workspace; create schema	Dev lead
Wed afternoon	Buyer and farmer forms; load seed data	Dev
Wed evening	PL/SQL matching function and results page	Dev lead
Wed evening	Print interview sheets; confirm market and transport	Field lead
Thu 6–10 AM	Market interviews	Field team
Thu morning	Map, grade routing, impact dashboard; deploy	Dev

When	Task	Owner
Thu by 11 AM	Share interview findings	Field team
Thu by 1 PM	Deck, README, 200-word description	Pitch lead
Thu by 3 PM	Full rehearsal; bug fixes	All
Thu by 3:30 PM	**Submit to ilabee@strathmore.edu (30-min buffer)**	Team lead
Thu evening	Two timed rehearsals; record backup demo video	All
Fri 8:00 AM	Arrive; laptop charged; demo open and logged in	All

10. Risks & Mitigations

Risk	Mitigation
OCI/APEX setup delays	Start immediately; ask Oracle technical mentors on Zoom; use apex.oracle.com workspace as fallback
Live demo fails on the day (Wi-Fi, session timeout)	Backup demo video and screenshots in the deck; phone hotspot
Too few interviews	Phone calls to farmer contacts from family or cooperatives as backup
Scope creep	Freeze features at Thursday 11:00 AM; extras go on the roadmap slide
Pitch runs over 7 minutes	Timed rehearsals; one speaker per slide; demo script

11. Open Decisions

1. Team members and roles: who leads development, field research and the pitch?
2. Does anyone already have OCI/APEX access? If not, set up today.
3. Naming: present as "HarvestLink by Bora Chakula", or choose one name?
4. Which market will the field team visit?

Appendix A: Interview Record Form

One form per interviewee. Tick consent before recording.

Interviewer / Date	
Market / Location	
Interviewee (first name, role)	☐ Farmer ☐ Trader ☐ Broker ☐ Other
Consent given	☐ Notes ☐ Photo ☐ Video
Crops & volume per season	
Buyer known before harvest?	☐ Yes ☐ No Notes:
% lost / spoiled, and why	
Transport: mode, cost, distance	
Broker price vs market price	
Lower-grade produce goes to	
Willing to aggregate?	☐ Yes ☐ Maybe ☐ No
Phone / M-Pesa / WhatsApp	☐ Smartphone ☐ Feature phone ☐ M-Pesa ☐ WhatsApp
Best quote	
Other notes	